

# CS492C: Diffusion and Flow Models

## Denoising Diffusion Probabilistic Models 1

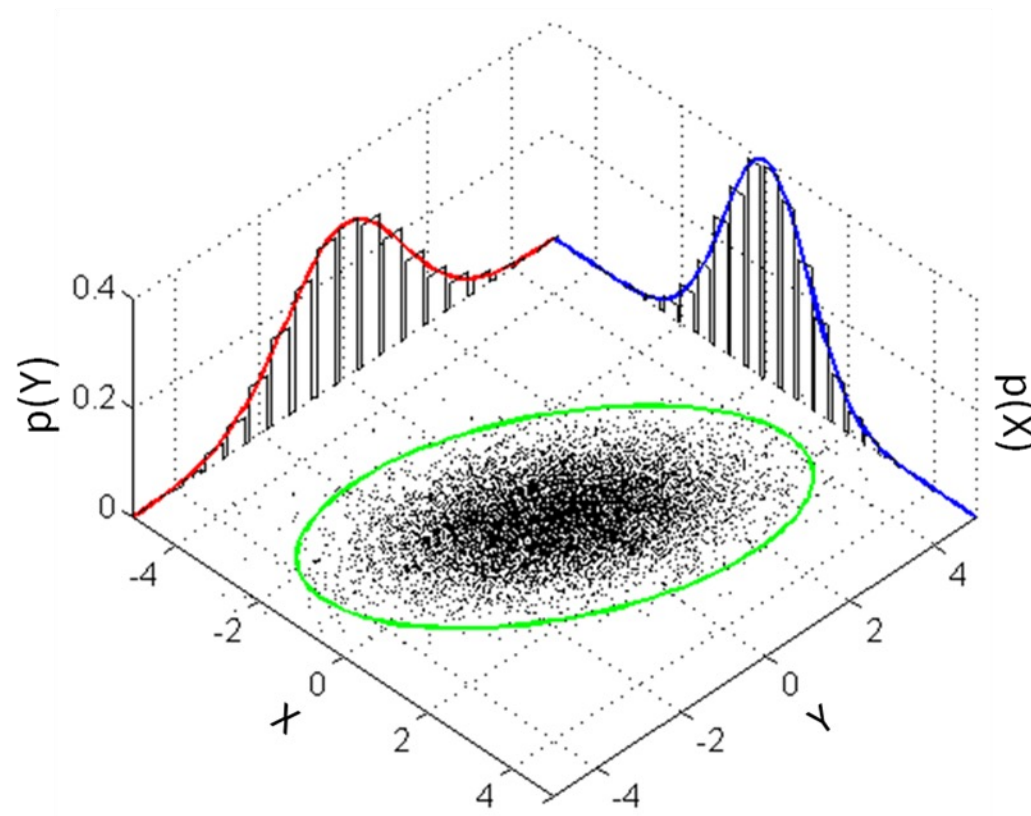
LECTURE 3  
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Fall 2025  
KAIST

# Previously in CS492C

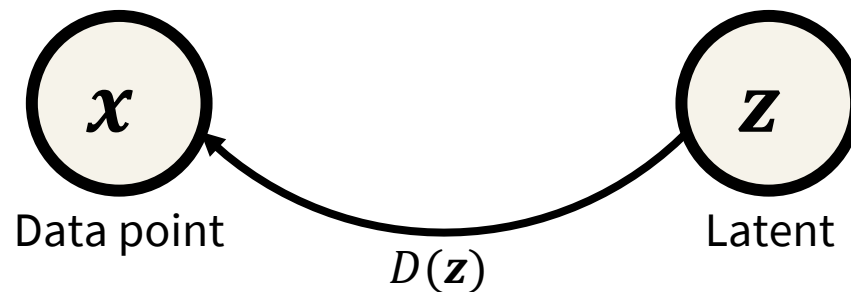
From a **statistical perspective**, we will view a dataset as

- there being a **probability distribution** of the data, and
- the given points are **samples** from the probability distribution.



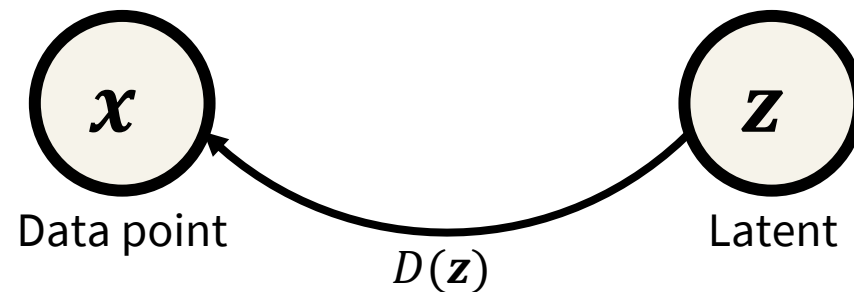
# Previously in CS492C

- **Map** a *simple* distribution  $p(\mathbf{z})$  (e.g., a standard normal distribution  $\mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$ ) to the **data distribution**  $p(\mathbf{x})$ .
  - $\mathbf{z}$ : **Latent** variable
  - $p(\mathbf{z})$ : **Latent** distribution
- Sample from  $p(\mathbf{z})$  and map it to a data point.



# Previously in CS492C

- How to map a latent distribution  $p(\mathbf{z})$  to the data distribution  $p(\mathbf{x})$  using a **neural network**?
- How to **guarantee** that a latent is mapped to a data point of the data distribution?
- We need an **additional** neural network.

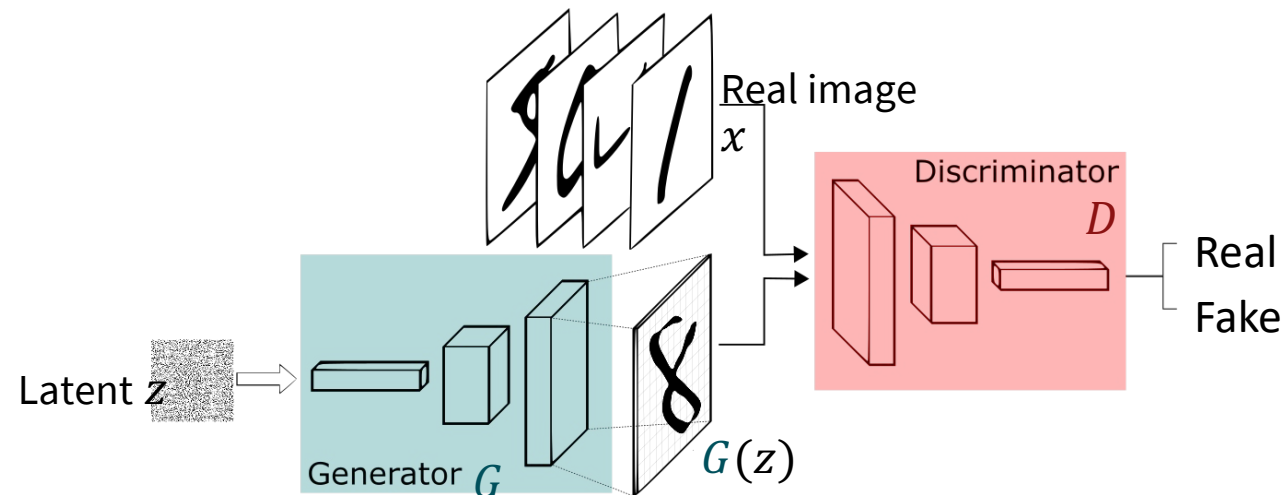


# Previously in CS492C

## Generative Adversarial Network (GAN)

Introduce a **discriminator** and have it compete with the generator (decoder).

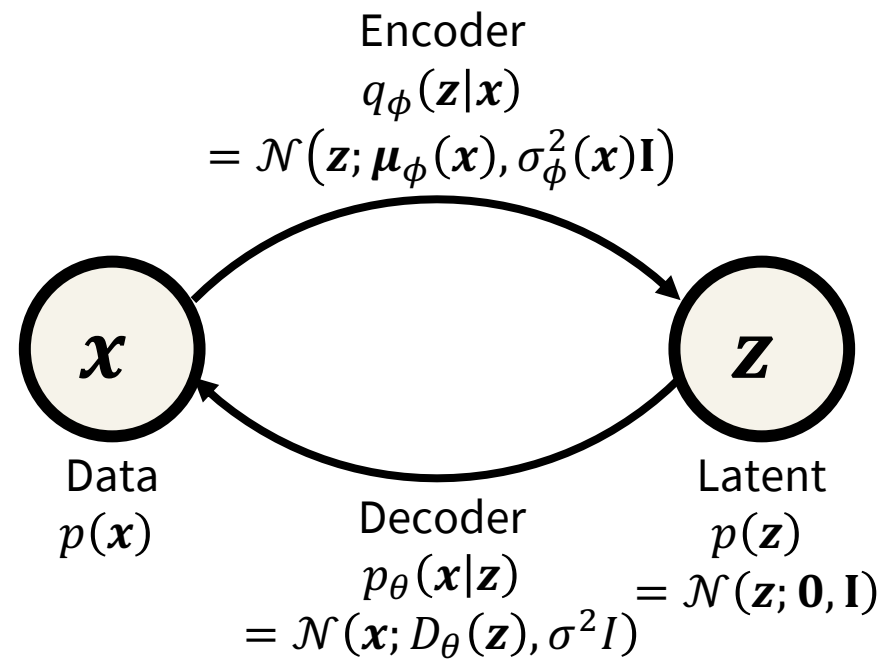
$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p(x)} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$



# Previously in CS492C

## Variational Autoencoder (VAE)

Introduce an **encoder** to learn a proxy of the posterior distribution.



# Previously in CS492C

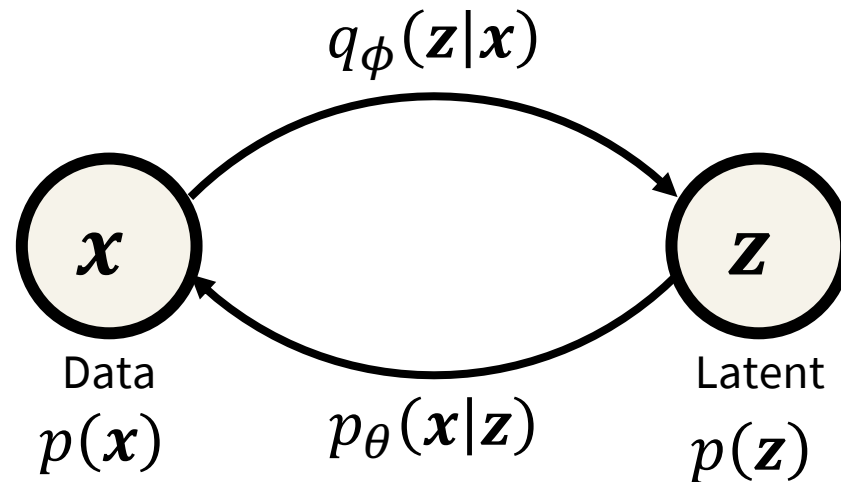
## Basics

- Marginal distribution
- Expected value
- Bayes' rule
- Kullback–Leibler (KL) Divergence
- Jensen's inequality

# Previously in CS492C

## Bayes' Rule

$$\boxed{p(\mathbf{z}|\mathbf{x})}^{\text{Posterior Encoder}} = \frac{\boxed{p(\mathbf{x}|\mathbf{z})}^{\text{Likelihood Decoder}} \boxed{p(\mathbf{z})}^{\text{Prior Latent}}}{\boxed{p(\mathbf{x})}^{\text{Marginal Data}}}$$





# Quiz

$$\begin{aligned} D_{\text{KL}}(p \parallel q) &= \int p(\mathbf{x}) \log \left( \frac{p(\mathbf{x})}{q(\mathbf{x})} \right) d\mathbf{x} = \mathbb{E}_{p(\mathbf{x})} \left[ \log \left( \frac{p(\mathbf{x})}{q(\mathbf{x})} \right) \right] \\ &= \mathbb{E}_{p(\mathbf{x})} [\log(p(\mathbf{x})) - \log(q(\mathbf{x}))] \end{aligned}$$

**Q.** When  $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \sigma^2 \mathbf{I})$  and  $q(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$ ,

What is  $D_{\text{KL}}(p \parallel q)$ ?

# Answer

If  $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_p, \sigma_p^2 \mathbf{I})$ ,  $q(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_q, \sigma_q^2 \mathbf{I})$ ,  $\mathbf{x} \in \mathbb{R}^d$ ,

$$D_{\text{KL}}(p \parallel q) = \frac{1}{2} \left[ d \left( \frac{\sigma_p^2}{\sigma_q^2} - 1 - \ln \frac{\sigma_p^2}{\sigma_q^2} \right) + \frac{\|\boldsymbol{\mu}_p - \boldsymbol{\mu}_q\|^2}{\sigma_q^2} \right].$$

When  $\boldsymbol{\mu}_q = 0$ ,  $\sigma_q = 1$ ,

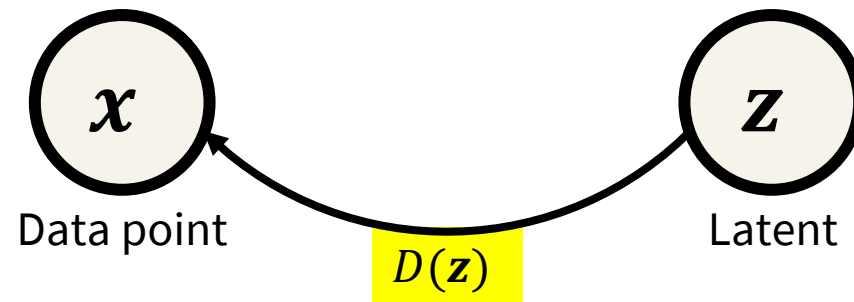
$$D_{\text{KL}}(p \parallel q) = \frac{1}{2} \left[ d(\sigma_p^2 - 1 - \ln \sigma_p^2) + \|\boldsymbol{\mu}_p\|^2 \right]$$

# Back to VAE...

In VAE, we want model  $p(\mathbf{x})$  as

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

where  $p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; D(\mathbf{z}), \sigma^2 I)$  and  $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$ .

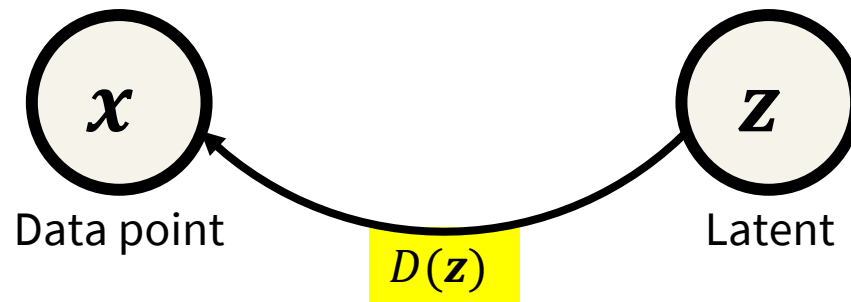


# Variational Autoencoder (VAE)

In VAE, we want model  $p(\mathbf{x})$  as

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_{\theta}(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$

where  $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; D_{\theta}(\mathbf{z}), \sigma^2 I)$  and  $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$ .



# Previously in CS492C

## Evidence Lower Bound (ELBO)

- $p(\mathbf{x})$  can be computed as  $\frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})}$ , but we cannot directly maximize it since  $p(\mathbf{z}|\mathbf{x})$  is unknown.
- Let's maximize **lower bound** of  $\log p(\mathbf{x})$ :

$$\log p(\mathbf{x}) \geq \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right]$$

# Previously in CS492C

$$\begin{aligned}\log p(\mathbf{x}) &= \log \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} \\ &= \log \int p(\mathbf{x}, \mathbf{z}) \frac{q_\phi(\mathbf{z}|\mathbf{x})}{q_\phi(\mathbf{z}|\mathbf{x})} d\mathbf{z} \\ &= \boxed{\log} \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[ \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \\ &\geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right]\end{aligned}$$

Concave  
function



# Evidence Lower Bound (ELBO)

Let's decompose ELBO:

$$\begin{aligned}\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] &= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \\&= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x}|\mathbf{z})] - \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[ \log \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} \right] \\&= \boxed{\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x}|\mathbf{z})]} - \boxed{D_{KL} \left( q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}) \right)}\end{aligned}$$

Reconstruction term  
to be *maximized*.

Prior matching term  
to be *minimized*.

# Evidence Lower Bound (ELBO)

**Q.** When does the Jensen gap

$$\left( \log p(\mathbf{x}) - \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \right)$$

become zero?



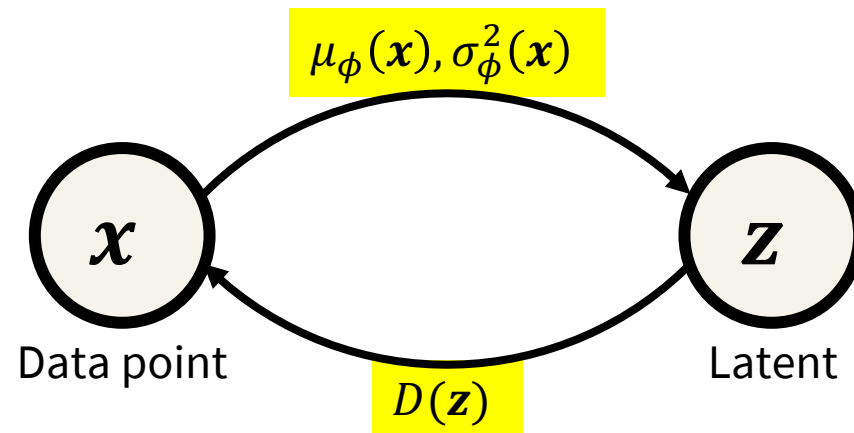
# Evidence Lower Bound (ELBO)

$$\begin{aligned} \mathbf{A.} \quad & \log p(\mathbf{x}) - \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \\ &= \log p(\mathbf{x}) - \left( \log p(\mathbf{x}) + \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[ \log \frac{p(\mathbf{z}|\mathbf{x})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \right). \\ &= D_{KL} \left( q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}|\mathbf{x}) \right). \end{aligned}$$

The **Jensen gap** becomes zero iff  $q_{\phi}(\mathbf{z}|\mathbf{x}) = p(\mathbf{z}|\mathbf{x})$  almost everywhere.

# Back to VAE...

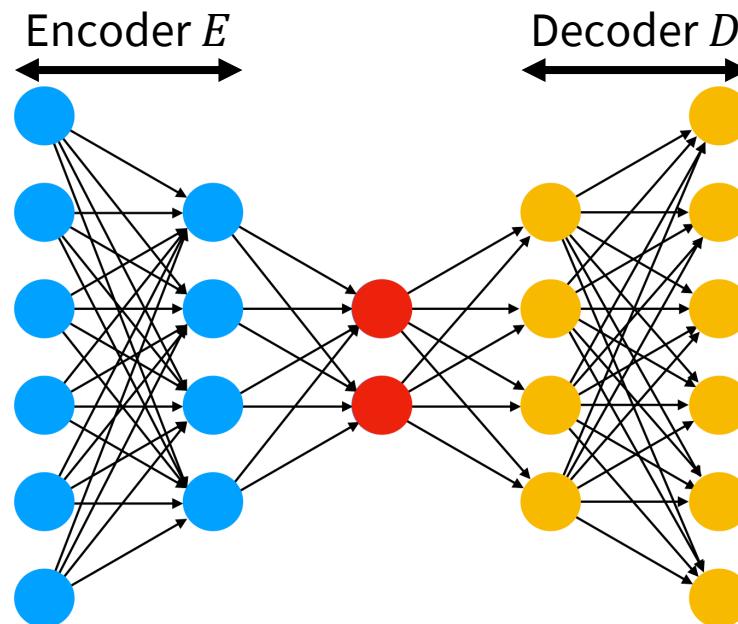
- How to maximize  $\log p(\mathbf{x})$ ? Maximize **ELBO**!
- We need the **proxy** distribution  $q_\phi(\mathbf{z}|\mathbf{x}) \rightarrow$  **Encoder**
- Let  $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_\phi(\mathbf{x}), \sigma_\phi^2(\mathbf{x})\mathbf{I})$



# Variational Autoencoder (VAE)

Same with autoencoder but,

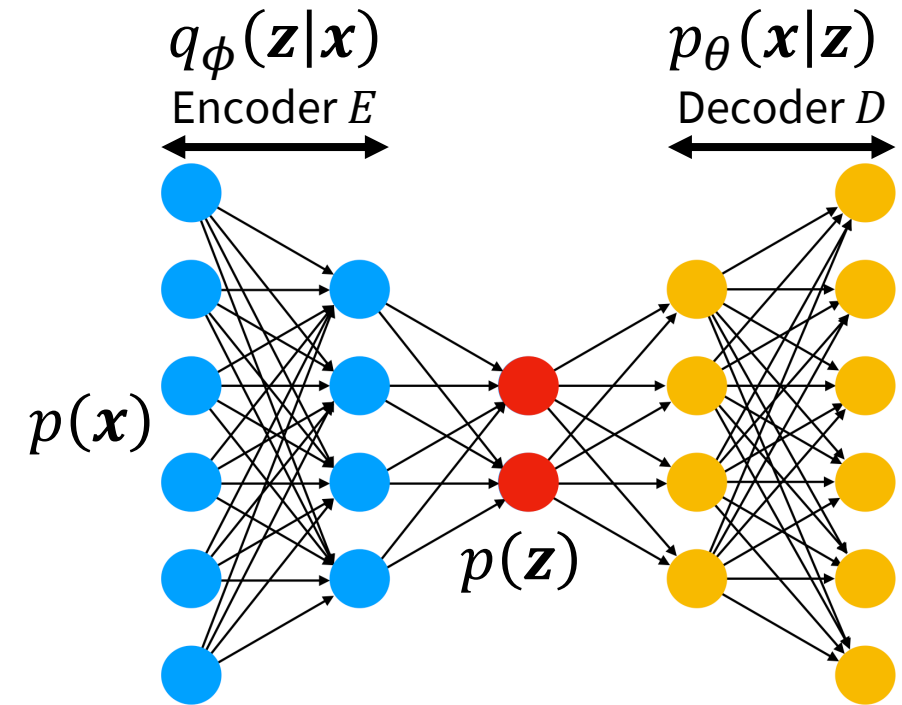
- From input  $\mathbf{x}$ , the **encoder** predicts  $\mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})$ .
- The **decoder** takes a sample  $\mathbf{z} \sim \mathcal{N}(\mathbf{z}; \mu_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})\mathbf{I})$  as input.



# Variational Autoencoder (VAE)

## Summary

|                     |                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Data distribution   | $p(\mathbf{x})$                                                                                                                         |
| Encoder             | $q_{\phi}(\mathbf{z} \mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_{\phi}(\mathbf{x}), \sigma_{\phi}^2(\mathbf{x})\mathbf{I})$ |
| Latent distribution | $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$                                                                       |
| Decoder             | $p_{\theta}(\mathbf{x} \mathbf{z}) = \mathcal{N}(\mathbf{x}; D_{\theta}(\mathbf{z}), \sigma^2\mathbf{I})$                               |



# Training

How to maximize **ELBO**?

$$\operatorname{argmax}_{\theta, \phi} \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}|\mathbf{z})] - D_{KL} \left( q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}) \right)$$

Approximates using a Monte Carlo estimate:

$$\operatorname{argmax}_{\theta, \phi} \frac{1}{N} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}|\mathbf{z}^{(i)}) - D_{KL} \left( q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}) \right)$$

where  $\mathbf{z}^{(i)} \sim q_{\phi}(\mathbf{z}|\mathbf{x})$  for the given  $\mathbf{x}$ .

# Training

How to maximize **ELBO**?

$$\operatorname{argmax}_{\theta, \phi} \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}|\mathbf{z})] - D_{KL} \left( q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}) \right)$$

Approximates using a Monte Carlo estimate:

$$\operatorname{argmax}_{\theta, \phi} \frac{1}{N} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}|\mathbf{z}^{(i)}) - D_{KL} \left( q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}) \right)$$

Reparameterization Trick

where  $\mathbf{z}^{(i)} = \boldsymbol{\mu}_{\phi}(\mathbf{x}) + \boldsymbol{\sigma}_{\phi}(\mathbf{x})\boldsymbol{\epsilon}$  where  $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathbf{I})$

# Training

Recall  $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; D_{\theta}(\mathbf{z}), \sigma^2 \mathbf{I})$ .

$$\log p_{\theta}(\mathbf{x}|\mathbf{z}^{(i)}) = \log \left( \frac{1}{\sqrt{(2\pi\sigma^2)^d}} \exp \left( -\frac{\|\mathbf{x} - D_{\theta}(\mathbf{z})\|^2}{2\sigma^2} \right) \right)$$

$$= -\frac{1}{2\sigma^2} \boxed{\|\mathbf{x} - D_{\theta}(\mathbf{z})\|^2} - \log \sqrt{(2\pi\sigma^2)^d}$$

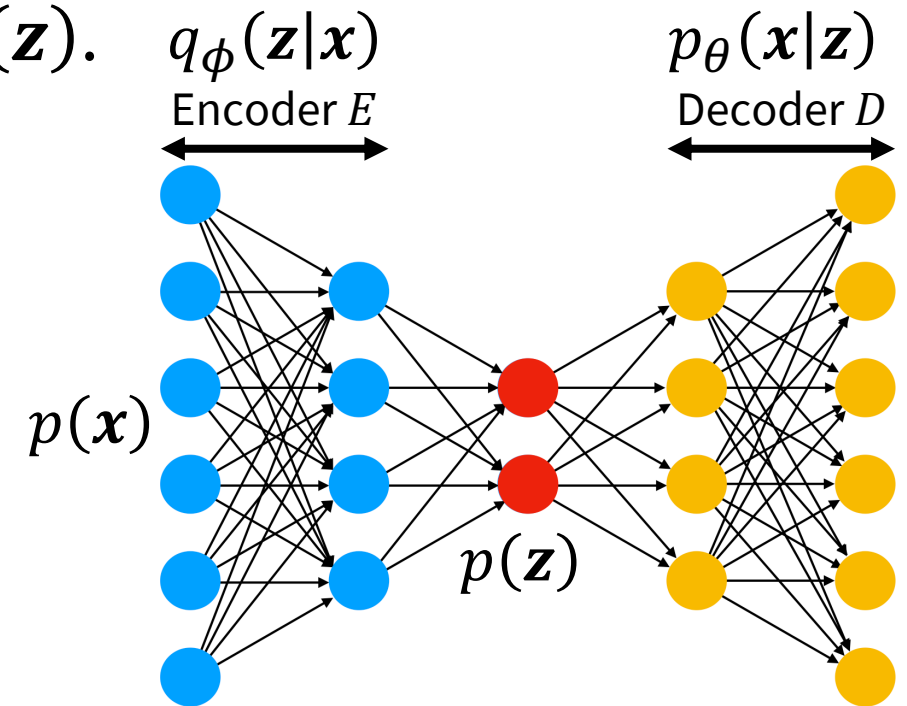
This is why it is called the reconstruction term.

Constant

# Training

1. Feed a data point  $\mathbf{x}$  to the encoder to predict  $\boldsymbol{\mu}_\phi(\mathbf{x})$  and  $\sigma_\phi^2(\mathbf{x})$ .
2. **Sample** a latent variable  $\mathbf{z}$  from  $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_\phi(\mathbf{x}), \sigma_\phi^2(\mathbf{x})\mathbf{I})$ .
3. Feed  $\mathbf{z}$  to the decoder to predict  $\hat{\mathbf{x}} = D_\theta(\mathbf{z})$ .
4. Compute the gradient decent through the negative ELBO.

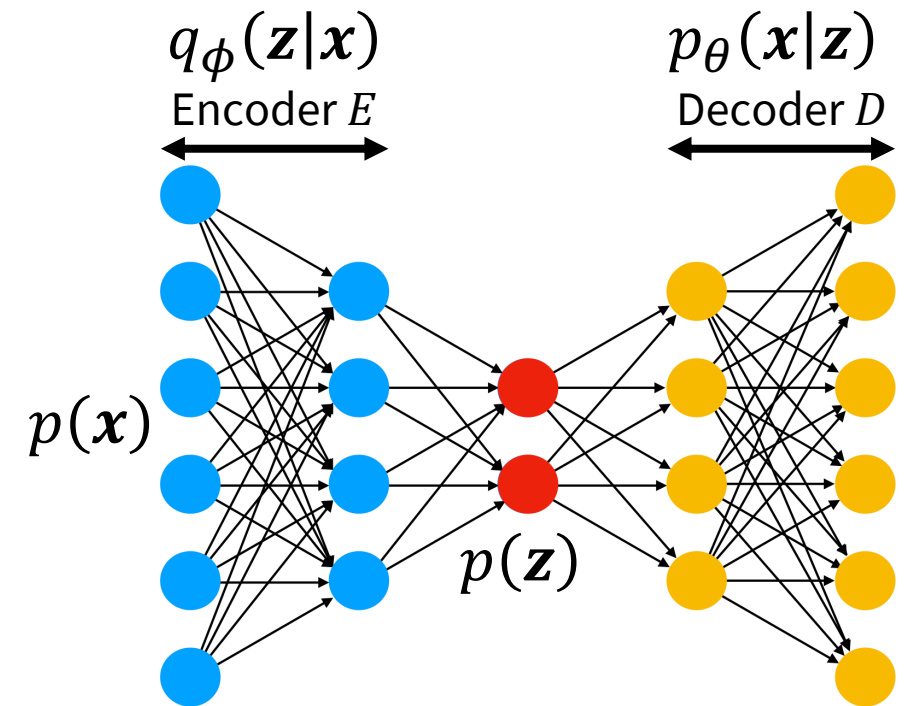
**Q.** Why is the **sampling** differentiable?





# Generation

1. Sample a latent variable  $\mathbf{z}$  from  $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$ .
2. Feed  $\mathbf{z}$  to the decoder to predict  $\hat{\mathbf{x}} = D_{\theta}(\mathbf{z})$ .



# Dimensions

**Q.** Should the **dimensions** of the input data and the latent variables be the same?

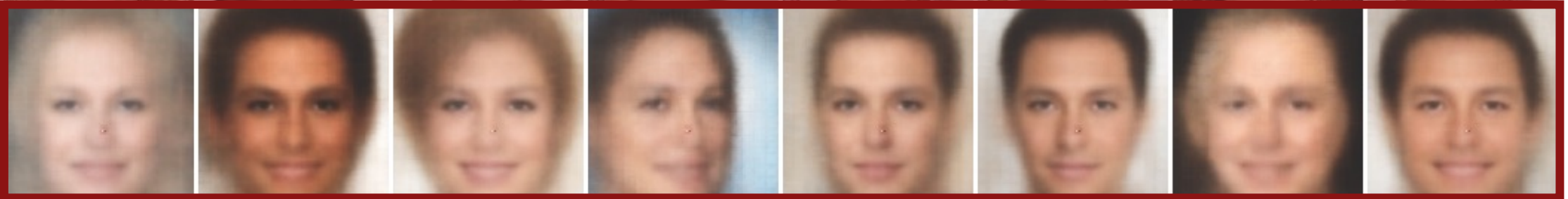
# Limitations of VAEs

## Typical failure cases of VAEs

Real  
images



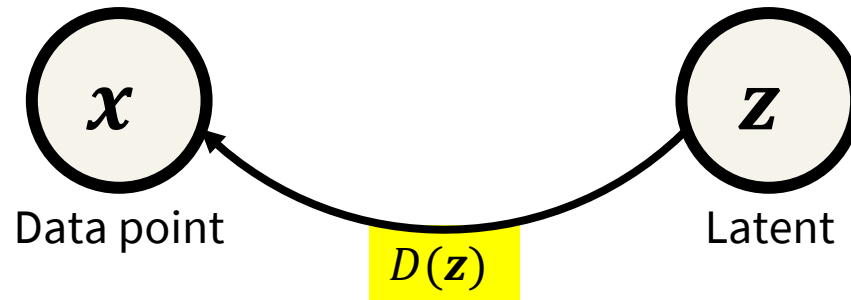
Generated  
Images



# Limitations of VAEs

Is a **Gaussian** distribution sufficient to model the likelihood distribution  $p_{\theta}(\mathbf{x}|\mathbf{z})$ ?

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_{\theta}(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$



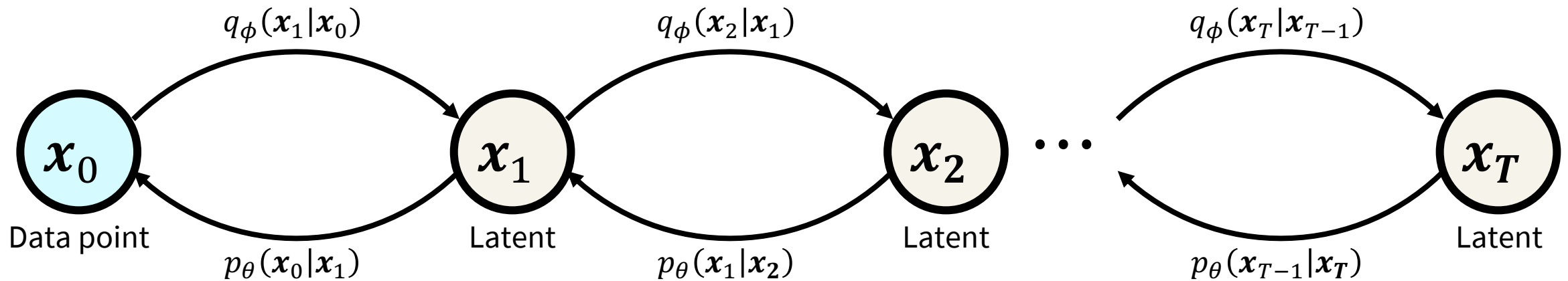
# VAE Variants

- Vector-Quantized VAE (VQ-VAE) [Oord et al., NeurIPS 2017]
- Beta-VAE [Higgins et al., ICLR 2017]
- Wasserstein VAE (WAE) [Tolstikhin et al., ICLR 2018]
- VAE-GAN [Larsen et al., ICML 2016]
- Normalizing Flow VAE [Rezende and Mohamed, ICML 2015]

# Hierarchical VAEs

Make a recursive (hierarchical) VAE.

- Data point:  $\mathbf{x} \rightarrow \mathbf{x}_0$
- Latent variable(s):  $\mathbf{z} \rightarrow \mathbf{x}_{1:T}$



# Markovian Hierarchical VAEs

Let's consider a **Markovian** process.

A Markov process is a stochastic process where the probability of each event **depends only on the previous state**.

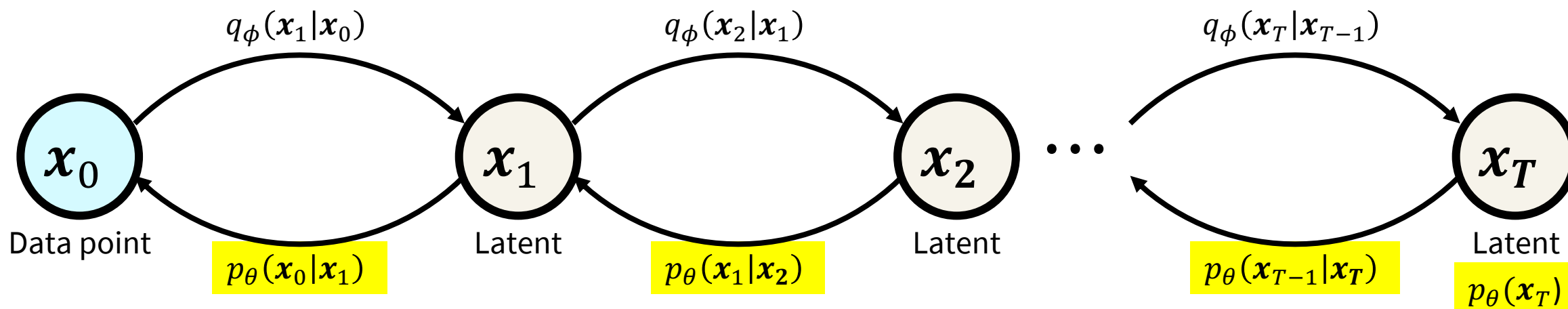
**“What happens next depends only on the state of affairs now!”**

# Markovian Hierarchical VAEs

Let's consider a Markovian process.

Joint distribution:

$$p_{\theta}(\mathbf{x}_{0:T}) = p_{\theta}(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$



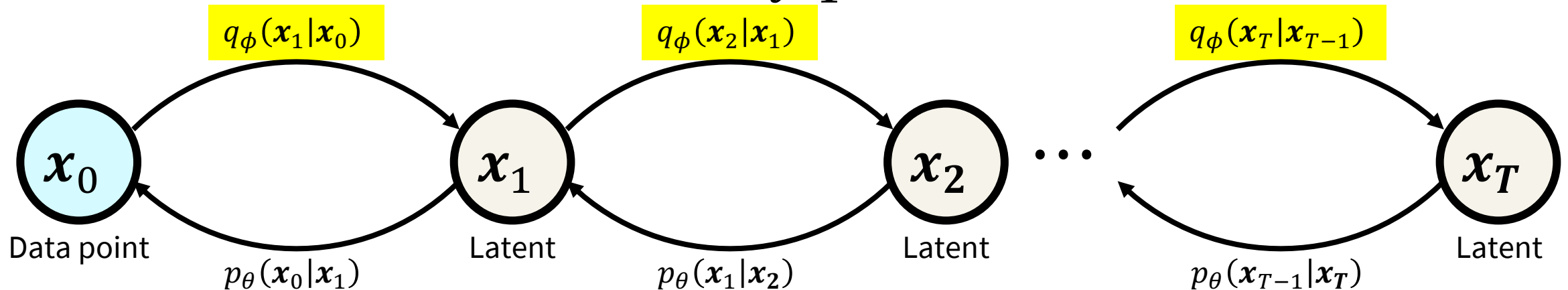


# Markovian Hierarchical VAEs

Let's consider a Markovian process.

Variational posterior:

$$q_{\phi}(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q_{\phi}(\mathbf{x}_t|\mathbf{x}_{t-1})$$



# Markovian Hierarchical VAEs

$$\begin{aligned}\log p(\mathbf{x}_0) &= \log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \\ &= \log \int p_\theta(\mathbf{x}_{0:T}) \frac{q_\phi(\mathbf{x}_{1:T} | \mathbf{x}_0)}{q_\phi(\mathbf{x}_{1:T} | \mathbf{x}_0)} d\mathbf{x}_{1:T} \\ &= \log \mathbb{E}_{q_\phi(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[ \frac{p_\theta(\mathbf{x}_{0:T})}{q_\phi(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right] \\ &\geq \mathbb{E}_{q_\phi(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[ \log \frac{p_\theta(\mathbf{x}_{0:T})}{q_\phi(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right]\end{aligned}$$

# Denoising Diffusion Probabilistic Models (DDPM)

Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020.

# Denoising Diffusion Probabilistic Models

Consider a special case of the Markovian hierarchical VAEs where:

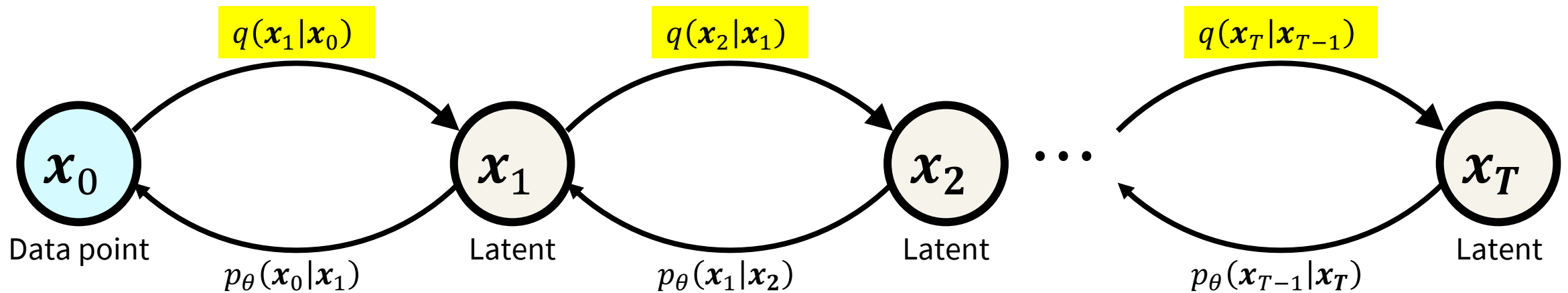
- the latent **dimension** is the **same** as the data dimension, and
- the **variational posteriors**  $q_{\phi}(\mathbf{x}_{t+1}|\mathbf{x}_t)$  are not learned but predefined:

$$q_{\phi}(\mathbf{x}_{t+1}|\mathbf{x}_t) \rightarrow \mathbf{q}(\mathbf{x}_{t+1}|\mathbf{x}_t)$$

# Terminology

**Forward** process (predefined):

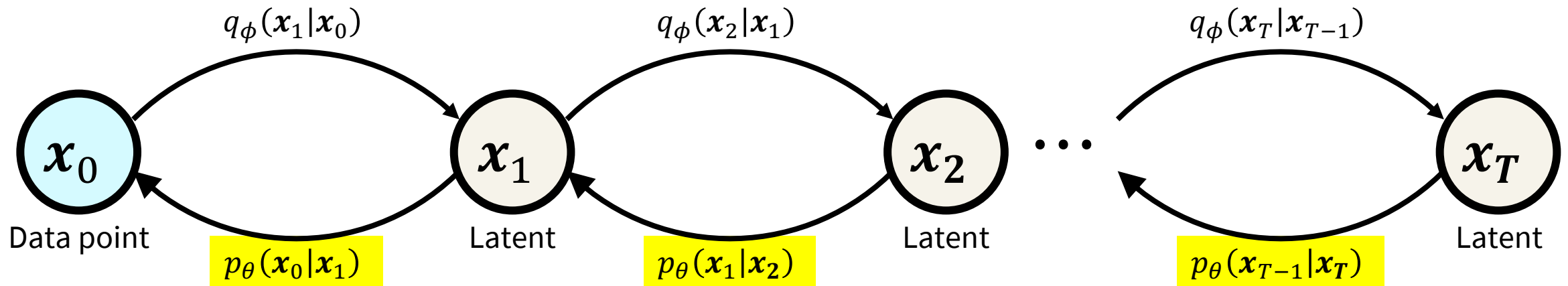
$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})$$



# Terminology

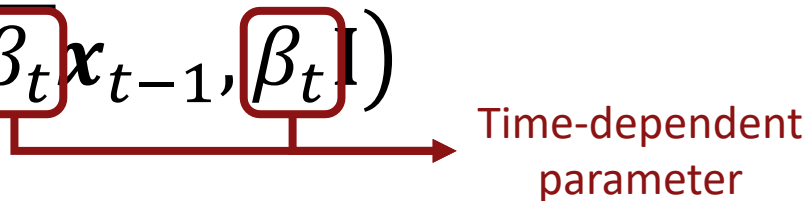
**Reverse** process (**learned**):

$$p_{\theta}(\mathbf{x}_0, \mathbf{x}_{1:T}) = p_{\theta}(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$$



## Forward Process (Data $\rightarrow$ Latent)

In the forward process, the transition distribution  $q(\mathbf{x}_t | \mathbf{x}_{t-1})$  is specifically predefined as:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$


Time-dependent parameter

where  $\{\beta_t \in (0, 1)\}_{t=1}^T$  and  $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$ .

*This is just one specific definition of forward transitions.  
We will see other formulations in later lectures.*

# Choice of $\beta_t$

- Learned.
- Constant.
- Linearly or quadratically increased.
- Follows a **cosine** function  
(Nichol and Dhariwal, Improved Denoising Diffusion Probabilistic Models, ICML 2021).
- Note that the reverse step  $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$  can be well approximated by a Gaussian form only when  $\beta_t$  is small ( $\beta_t \ll 1$ ).



# How to maximize ELBO in this case?

Disclaimer: We'll skip some complicated equations in the following slides.

# ELBO

How to minimize the *negative* ELBO in this case?

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \leq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[ -\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$$= \dots =$$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

Reconstruction  
term

$$+\mathbb{E}_{q(\mathbf{x}_{T-1}|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_T|\mathbf{x}_{T-1}) \parallel p(\mathbf{x}_T))]$$

Prior matching  
term

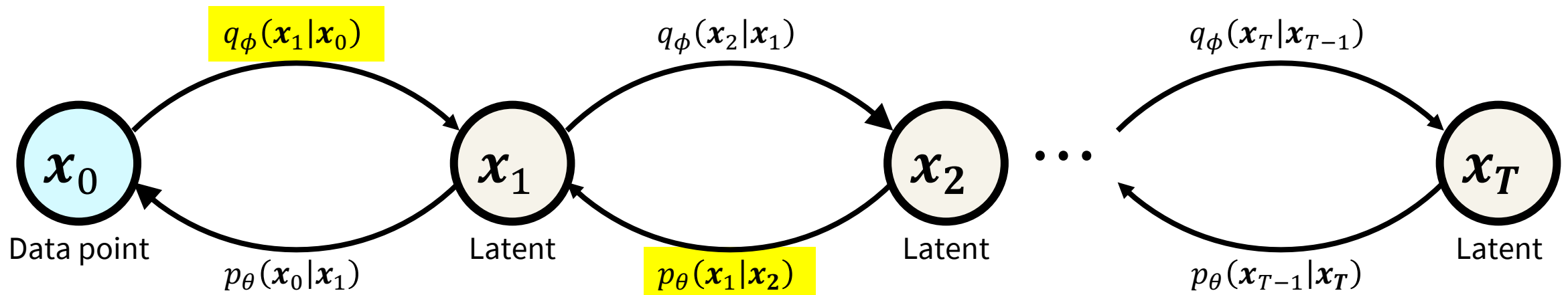
$$+\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1}|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_t|\mathbf{x}_{t-1}) \parallel p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1}))]$$

Consistency  
term

# Consistency Term

$$\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1} | \mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_t | \mathbf{x}_{t-1}) \parallel p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1}))]$$

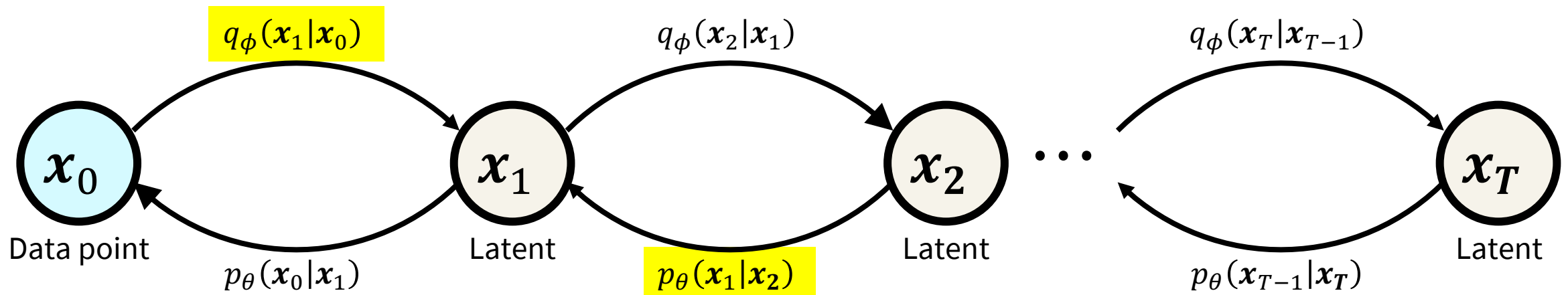
Make the forward and reverse steps be consistent at each time step.



# Consistency Term

$$\sum_{t=1}^{T-1} \mathbb{E}_{q(\mathbf{x}_{t-1}, \mathbf{x}_{t+1} | \mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_t | \mathbf{x}_{t-1}) \parallel p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t+1}))]$$

Expectation over two random variables; computationally expensive.



# ELBO

Can we avoid having two random variables in an expectation?

Let's re-decompose the ELBO using the fact that

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0).$$

**Q.** Why  $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0)$ ?

# ELBO

Decompose the negative ELBO in a **different** way :

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \leq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[ -\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$$= \dots =$$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

Reconstruction  
term  $\mathcal{L}_0$

$$+ D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))$$

New prior matching  
term  $\mathcal{L}_T$

$$+ \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Denoising  
matching  
term  $\mathcal{L}_{t-1}$

# Reconstruction Term $\mathcal{L}_0$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}[\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)]$$

The same loss term as in VAE, but applied only to the **final** reverse step.

# ELBO

Decompose the negative ELBO in a **different** way :

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \leq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[ -\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$= \dots =$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

$$+ D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))$$

New prior matching  
term  $\mathcal{L}_T$

$$+ \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$



# Prior Matching Term $\mathcal{L}_T$

# Prior Matching Term $\mathcal{L}_T$

$$D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))$$

- Identical to the KL divergence term in VAE.
- Note that there is nothing to be optimized;  $q(\mathbf{x}_T|\mathbf{x}_0)$  and  $p(\mathbf{x}_T)$  are predefined.

# Forward Convergence

Then,  $q(\mathbf{x}_T | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$ ?

**Yes, under certain assumptions.**

Recall

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

where  $\{\beta_t \in (0, 1)\}_{t=1}^T$  and  $\beta_1 < \beta_2 < \dots < \beta_T$ .

# Forward Convergence

Then,  $q(\mathbf{x}_T | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$ ?

Yes, under certain assumptions.

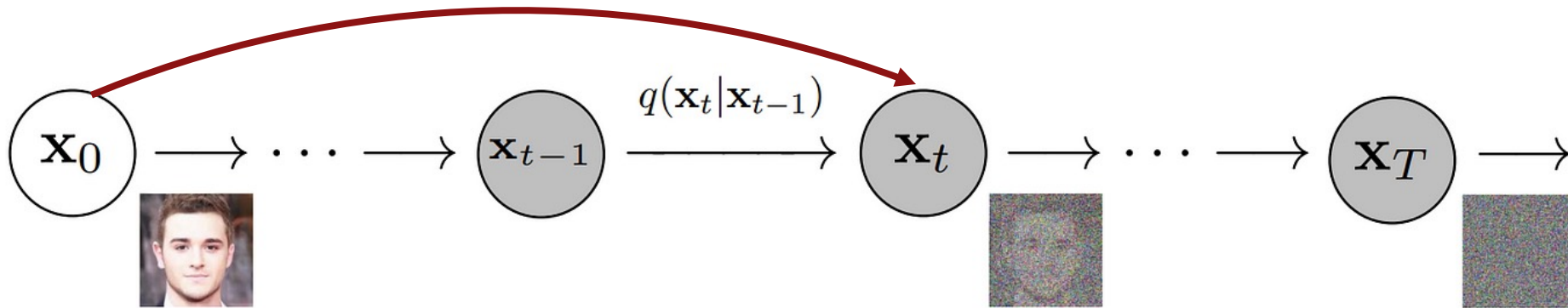
Let  $\alpha_t = 1 - \beta_t$ .

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

where  $\{\alpha_t \in (0, 1)\}_{t=1}^T$  and  $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_T$ .

$$q(\mathbf{x}_t | \mathbf{x}_0)$$

Can we derive  $q(\mathbf{x}_t | \mathbf{x}_0)$  from the sequence of  $q(\mathbf{x}_{t'} | \mathbf{x}_{t'-1})$   
for  $t = 1, \dots, t'$ ?



# Basics: Combination of Gaussian Variables

Suppose  $\mathbf{x}_1 \sim \mathcal{N}(\mu_1, \sigma_1^2 \mathbf{I})$  and  $\mathbf{x}_2 \sim \mathcal{N}(\mu_2, \sigma_2^2 \mathbf{I})$ .

**Q.** What is the distribution of  $\mathbf{x}_1 + \mathbf{x}_2$ ?

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**A.**  $\mathbf{x}_1 + \mathbf{x}_2 \sim \mathcal{N}(\mu_1 + \mu_2, (\sigma_1^2 + \sigma_2^2) \mathbf{I})$

# Basics: Combination of Gaussian Variables

Suppose  $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ , and

$$\boldsymbol{x}_1 = \sigma_1 \boldsymbol{\varepsilon}_1 \text{ and } \boldsymbol{x}_2 = \sigma_2 \boldsymbol{\varepsilon}_2.$$

**Q.** What is the distribution of  $\boldsymbol{x}_1 + \boldsymbol{x}_2$ ?



# Basics: Combination of Gaussian Variables

Suppose  $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ , and

$$\boldsymbol{x}_1 = \sigma_1 \boldsymbol{\varepsilon}_1 \text{ and } \boldsymbol{x}_2 = \sigma_2 \boldsymbol{\varepsilon}_2.$$

$$\mathbf{A. } \boldsymbol{x}_1 + \boldsymbol{x}_2 \sim \mathcal{N}(\mathbf{0}, (\sigma_1^2 + \sigma_2^2)\mathbf{I}).$$

$\boldsymbol{x}_1 + \boldsymbol{x}_2 = \sqrt{\sigma_1^2 + \sigma_2^2} \boldsymbol{\varepsilon}$  where  $\boldsymbol{\varepsilon}$  is another standard normal sample.

# Forward Convergence

$$q(\mathbf{x}_1|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_1; \sqrt{\alpha_1}\mathbf{x}_0, (1 - \alpha_1)\mathbf{I})$$

$$q(\mathbf{x}_2|\mathbf{x}_1) = \mathcal{N}(\mathbf{x}_2; \sqrt{\alpha_2}\mathbf{x}_1, (1 - \alpha_2)\mathbf{I})$$

**Q.** What is the distribution of  $q(\mathbf{x}_2|\mathbf{x}_0)$ ?

**Hint.** Let's use the **reparamaterization trick**:

$$\mathbf{x}_1 = \sqrt{\alpha_1}\mathbf{x}_0 + \sqrt{1 - \alpha_1}\boldsymbol{\epsilon}_0$$

$$\mathbf{x}_2 = \sqrt{\alpha_2}\mathbf{x}_1 + \sqrt{1 - \alpha_2}\boldsymbol{\epsilon}_1$$

$$\boldsymbol{\epsilon}_0, \boldsymbol{\epsilon}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

# Forward Convergence

$$\mathbf{A.} \quad \mathbf{x}_2 = \sqrt{\alpha_2} \mathbf{x}_1 + \sqrt{1 - \alpha_2} \boldsymbol{\epsilon}_1$$

$$= \sqrt{\alpha_2} (\sqrt{\alpha_1} \mathbf{x}_0 + \sqrt{1 - \alpha_1} \boldsymbol{\epsilon}_0) + \sqrt{1 - \alpha_2} \boldsymbol{\epsilon}_1$$

$$= \sqrt{\alpha_2 \alpha_1} \mathbf{x}_0 + \sqrt{\alpha_2 (1 - \alpha_1)} \boldsymbol{\epsilon}_0 + \sqrt{1 - \alpha_2} \boldsymbol{\epsilon}_1$$

$$= \sqrt{\alpha_2 \alpha_1} \mathbf{x}_0 + \sqrt{(1 - \alpha_2 \alpha_1)} \bar{\boldsymbol{\epsilon}}_0$$

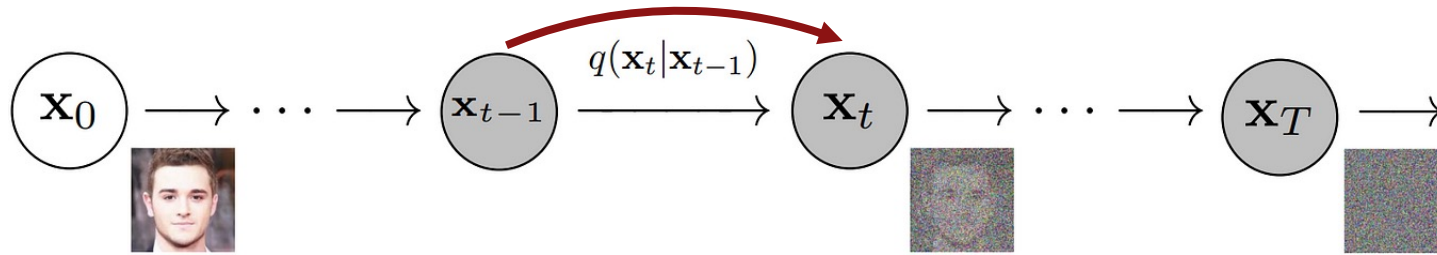
$$\therefore q(\mathbf{x}_2 | \mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_2 \alpha_1} \mathbf{x}_1, (1 - \alpha_2 \alpha_1) \mathbf{I})$$

# Forward Convergence

$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\ &= \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \boldsymbol{\epsilon}_{t-2}) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \boldsymbol{\epsilon}_{t-2} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{(1 - \alpha_t \alpha_{t-1})} \bar{\boldsymbol{\epsilon}}_{t-2} \\ &= \dots \\ &= \sqrt{\prod_{i=1}^t \alpha_i} \mathbf{x}_0 + \sqrt{(1 - \prod_{i=1}^t \alpha_i)} \bar{\boldsymbol{\epsilon}}_0 \end{aligned}$$

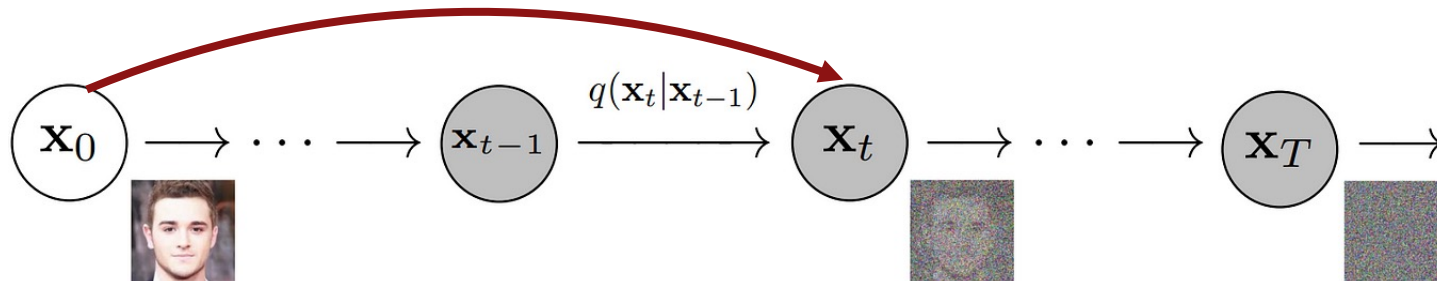
# Forward Convergence

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$



$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

where  $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$  *Also a normal distribution!*

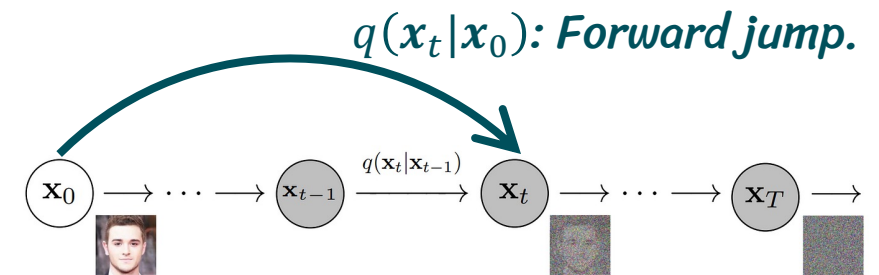


$$q(\mathbf{x}_t | \mathbf{x}_0)$$

Given  $\mathbf{x}_0$ ,  $\mathbf{x}_t$  at any arbitrary timestep  $t$  can be directly sampled from a Gaussian distribution without a Markov chain:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}).$$

Note that  $\bar{\alpha}_1 > \bar{\alpha}_2 > \dots > \bar{\alpha}_T$ .



# Forward Convergence

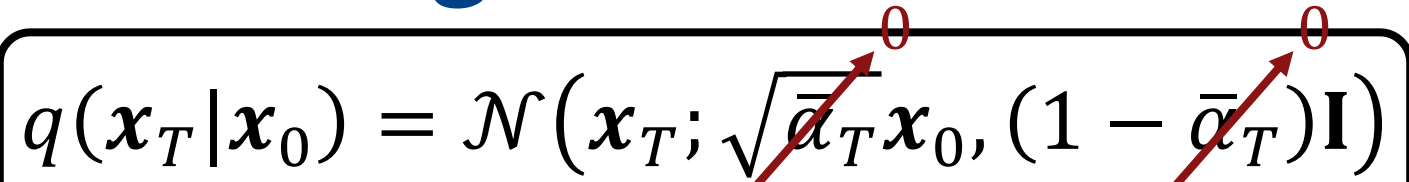
$$q(\mathbf{x}_T | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_T; \sqrt{\bar{\alpha}_T} \mathbf{x}_0, (1 - \bar{\alpha}_T) \mathbf{I})$$

where  $\bar{\alpha}_T = \prod_{t=1}^T (1 - \beta_t)$ .

**Q.** When  $\{\beta_t \in (0, 1)\}_{t=1}^T$ , What is

$$\lim_{T \rightarrow \infty} \bar{\alpha}_T = \lim_{T \rightarrow \infty} \prod_{t=1}^T (1 - \beta_t) ?$$

# Forward Convergence

$$q(\mathbf{x}_T | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_T; \sqrt{\bar{\alpha}_T} \mathbf{x}_0, (1 - \bar{\alpha}_T) \mathbf{I})$$


where  $\bar{\alpha}_T = \prod_{t=1}^T (1 - \beta_t)$ .

As  $T \rightarrow \infty$ ,  $q(\mathbf{x}_T | \mathbf{x}_0)$  converges to the standard normal distribution  $\mathcal{N}(\mathbf{x}; \mathbf{0}, \mathbf{I})$ .



# Prior Matching Term $\mathcal{L}_T$

$$D_{KL}(q(\mathbf{x}_T | \mathbf{x}_0) \parallel p(\mathbf{x}_T))$$

Close to zero by the definition of the forward transition distribution  $q(\mathbf{x}_t | \mathbf{x}_{t-1})$ . *Nothing to do for the optimization.*